Unsupervised Learning

1. A clustering algorithm produces clusters, but the resulting clusters do not correspond to any meaningful real-world groups. What is the BEST conclusion?

A. The algorithm is incorrect
B. The dataset must be supervised
C. Clustering does not guarantee meaningful semantic groups
D. Increasing the number of clusters will always solve the problem

2. Which statement about unsupervised learning is TRUE?

A. It always requires ground-truth labels during training
B. It discovers patterns or structure without predefined target labels
C. It can only be used for clustering
D. It always produces interpretable results

3. A dataset contains 1000 customers and 20 features, but no labels. You apply K-Means with K=4. What does K-Means actually learn?

A. The four correct customer categories
B. Four predefined customer classes
C. Four groups based on similarity according to the chosen representation and distance
D. Four classes with maximum classification accuracy

4. An unsupervised algorithm identifies three clusters. Can we conclude that the dataset naturally contains exactly three groups?

A. Yes
B. No
C. Yes, if SSE is low
D. Yes, if K-Means converges

5. Which statement is MOST accurate about clustering and classification?

A. Clustering and classification are essentially the same
B. Classification discovers labels, while clustering predicts known labels
C. Classification uses known target labels, while clustering generally does not
D. Both require labeled training data

6. A clustering algorithm gives different cluster assignments every time it runs. Which is the MOST likely explanation?

A. Unsupervised learning cannot be deterministic
B. Random initialization or stochastic optimization may be involved
C. The dataset must contain labels
D. Clustering algorithms cannot converge

7. Consider the following statement:

"Since K-Means minimizes SSE, the solution with the smallest SSE is always the most meaningful clustering."

What is wrong with this statement?

A. K-Means does not minimize SSE
B. SSE cannot be calculated for clusters
C. A lower SSE does not necessarily imply more meaningful or useful clusters
D. SSE is only used in supervised learning

8. Suppose K-Means produces an SSE of 100 for K=3 and 50 for K=6. Can we conclude that K=6 is better?

A. Yes
B. No, because SSE generally decreases as K increases
C. Yes, because smaller SSE always means better clustering
D. No, because SSE should increase with K

9. Why is the Elbow Method used in clustering?

A. To guarantee the globally optimal K
B. To identify a point where increasing K provides diminishing improvement
C. To maximize classification accuracy
D. To remove outliers

10. A dataset contains two well-separated groups, but one group is much larger than the other. Which statement is TRUE?

A. K-Means is guaranteed to identify both groups correctly
B. Cluster size imbalance can affect the resulting K-Means solution
C. K-Means automatically balances cluster sizes
D. Cluster size has no effect on distance-based clustering

11. PCA is applied before clustering. Which statement is TRUE?

A. PCA converts an unsupervised problem into a supervised problem
B. PCA always improves clustering accuracy
C. PCA may remove dimensions that contain useful clustering information
D. PCA requires class labels

12. A clustering algorithm produces very compact clusters. Does this necessarily mean the clustering is good?

A. Yes
B. No; clusters should also be sufficiently separated and meaningful
C. Yes, compactness is the only criterion
D. No, because compact clusters are undesirable

13. Which situation is LEAST suitable for standard K-Means?

A. Compact spherical clusters
B. Numerical features with similar scales
C. Clusters with strongly non-spherical shapes
D. Relatively well-separated groups

14. Suppose all features are measured in different units, such as age in years and income in lakhs. You directly apply K-Means. What is the primary concern?

A. K-Means cannot process multiple features
B. Income may dominate the distance calculation
C. Age will always dominate
D. K-Means automatically normalizes all features

15. Which statement about outliers in K-Means is MOST accurate?

A. K-Means automatically removes outliers
B. Outliers can significantly shift centroids
C. Outliers have no effect because K-Means uses medians
D. Outliers always form their own cluster

16. A clustering algorithm assigns every point to exactly one cluster. Which assumption does this describe?

A. Hard clustering
B. Soft clustering
C. Hierarchical clustering
D. Dimensionality reduction

17. In soft clustering, a data point can:

A. Belong to multiple clusters with different degrees of membership
B. Belong to exactly one cluster
C. Never belong to a cluster
D. Only belong to the largest cluster

18. A researcher clusters patients using their medical measurements and then observes that one cluster has a high disease rate. Can the researcher immediately claim that clustering discovered a disease category?

A. Yes
B. No; the association should be evaluated and validated independently
C. Yes, because clustering discovers hidden labels
D. Yes, if K-Means was used

19. Which statement about unsupervised learning is FALSE?

A. It can be used for exploratory data analysis
B. It can identify previously unknown patterns
C. It always discovers objectively correct groups

20. A clustering algorithm perfectly separates the dataset into two groups. However, after changing the feature scaling, the clustering changes completely. What does this MOST strongly indicate?

A. One of the algorithms must be wrong
B. The clustering is dependent on the representation and distance metric
C. Unsupervised learning cannot be used on numerical data
D. The original clustering was necessarily incorrect